

Orders of reaction and reaction mechanism (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R2.2.10) Identify the number of particles in the rate determining step from the order with respect to a reactant and deduce the overall order of reaction.
- (R2.2.10) Sketch, identify and analyse graphical representations of zero order, first order and second order reactions.
- (R2.2.11) Solve problems using the rate equation including finding the units of k .

In chemical reactions, changing the concentration of a reactant typically results in a change in the rate. However, not all reactants have the same effect on the rate. Experimental evidence shows that some reactants have a greater effect on the rate than other reactants and there are situations where changing the concentration of one of the reactants has no effect on the rate at all – how can this be? How does your understanding of reaction mechanisms and molecularity support the experimental evidence of rates of reaction?

Orders of reaction and their graphical representation

For a reaction between A and B the powers to which the concentrations of the reactants are raised are called the orders of reaction with respect to A and B. The orders of reaction you are going to deduce are 0, 1 and 2. An order of 3 is also possible but rarely seen. For example, if the rate of reaction is directly proportional to the concentration of the reactant A (that is the concentration of A is raised to the power of 1) then the order of reaction with respect to A is 1. Similarly, if the rate of reaction is proportional to the square of the concentration of B then the order of reaction with respect to B is 2.

When the concentration of a reactant does not affect the rate, the order of reaction with respect to this reactant is 0 as the concentration of the reactant is raised to the power of zero. Since anything raised to the power of zero is 1, then the concentration of this reactant will not appear in the rate expression.

The overall order of the reaction is expressed by adding the individual orders of each reactant. So if the reaction is first order with respect to A, that is, $\text{rate} \propto [\text{A}]$, and second order with respect to B, that is, $\text{rate} \propto [\text{B}]^2$, then the overall order of reaction is three ($1 + 2 = 3$).

The third order rate equation can be expressed as:

$$\text{rate} \propto [\text{A}] [\text{B}]^2$$

introducing the proportionality constant gives:

$$\text{rate} = k [\text{A}] [\text{B}]^2$$

The proportionality constant is known as the rate constant, k . It is constant for each reaction given that the only possible changes are to the concentrations of the reactants. The effect of temperature, catalyst and activation energy on the rate constant will be discussed in [section R2.2.12–13](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/arrhenius-equation-hl-id-45513/>).

Study skills

The rate constant is denoted by the small, italicised letter k .

You can use a concentration–time graph, see [section R2.2.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-of-reaction-id-43481/>), to see how the concentration of any reactant changes over time and how it affects the rate of the reaction. However, it is very difficult to differentiate between first order and second order graphs. A better picture of the effect of concentration on the rate can be obtained using rate-concentration graphs.

Zero order reaction

Changing the concentration of a reactant A does not affect the rate of reaction. The reaction is said to be zero order with respect to A.

A zero order reaction can be represented graphically as:

- Concentration—time graph: a straight line is obtained with a negative gradient.
- Rate—concentration graph: a horizontal line is produced representing a constant rate.

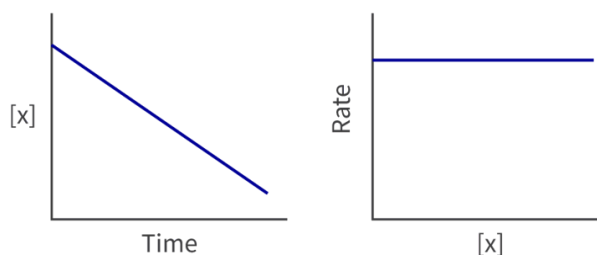


Figure 1. A concentration—time graph and a rate—concentration graph for a zero order reaction.

 More information for figure 1

The image displays two graphs side by side. The left graph is a concentration–time graph for a zero order reaction. The X-axis represents time, and the Y-axis represents concentration $[x]$. The line on this graph is a straight diagonal decreasing from the top left to the bottom right, indicating that concentration decreases linearly over time. The right graph is a rate–concentration graph for a zero order reaction. The X-axis represents concentration $[x]$, and the Y-axis represents the rate. This graph shows a horizontal line, indicating that the rate remains constant regardless of the concentration.

[Generated by AI]

First order reaction

When the concentration of a reactant A is raised to the power of 1, it means the concentration is directly proportional to the rate. The reaction is first order with respect to A and the rate equation can be written as $\text{rate} = k[A]$.

The overall order of reaction is also 1 in this case.

A first order reaction can be represented graphically as:

- Concentration—time graph: a curve showing a decreasing rate of reaction.
- Rate—concentration graph: a straight line passing through the origin with a positive gradient.

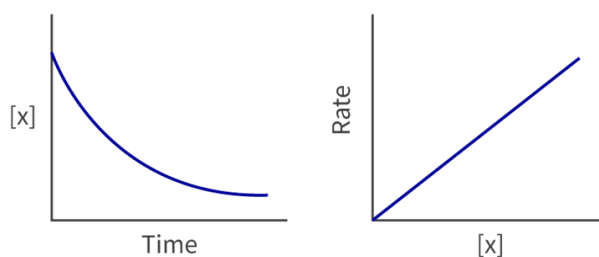


Figure 2. A concentration—time graph and a rate—concentration graph for a first order reaction.

 More information for figure 2

The image contains two separate graphs. The first graph on the left is a concentration-time graph. The X-axis represents time, with no specific values labeled, indicating a continuous progression. The Y-axis represents concentration, labeled as '[x]'. The graph shows a curve that decreases over time, depicting the diminishing concentration of a reactant in a first order reaction as time progresses.

The second graph on the right is a rate-concentration graph. The X-axis is labeled with '[x]', representing concentration, and the Y-axis is labeled 'Rate'. This graph illustrates a straight line with a positive slope, showing that the rate of reaction increases linearly with an increase in concentration in a first order reaction. Each axis is clearly labeled to represent the relationship between concentration and rate.

[Generated by AI]

Second order reactions

A reaction is said to be second order with respect to reactant A if the concentration of A is raised to the power of two. This means doubling the concentration of A increases the rate four times. The rate equation can be written as $\text{rate} = k[A]^2$.

A second order reaction can be represented graphically as:

- Concentration—time graph: as the concentration decreases the rate also decreases, in the form of a negative curve that slopes downwards.
- Rate—concentration graph: a curve is produced for this as well, but this curve is a positive curve, sloping upwards.

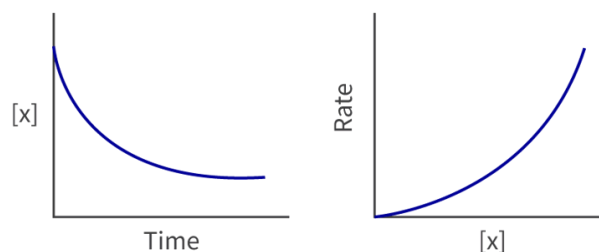


Figure 3. A concentration—time graph and a rate—concentration graph for a second order reaction.

 More information for figure 3

The image features two graphs. The first graph on the left is a concentration versus time graph for a second order reaction. The X-axis is labeled 'Time' and the Y-axis is labeled '[x]', representing concentration. The curve on this graph shows a downward slope, representing a decrease in concentration over time.

The second graph on the right is a rate versus concentration graph. The X-axis is labeled '[x]' indicating concentration, and the Y-axis is labeled 'Rate'. The curve on this graph slopes upward, indicating the rate increases with concentration.

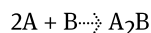
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Study skills

Keep in mind, the order of reaction can only be determined experimentally. You cannot deduce anything about the mechanism or order of reaction by looking at the reaction equation.

Worked example 1

A number of rate expressions have been given for the reaction



For each of the rate expressions identify the rate order with respect to each reactant and also identify the overall order of reaction.

1. rate = $k[A][B]$
2. rate = $k[A]^2[B]$
3. rate = $k[A][B]^2$
4. rate = $k[B]^2$

1. The reaction is first order with respect to A and first order with respect to B. The overall order of reaction is 2 ($1 + 1 = 2$).
2. The reaction is second order with respect to A and first order with respect to B. The overall order of reaction is 3 ($2 + 1 = 3$).
3. The reaction is first order with respect to A and second order with respect to B. The overall order of reaction is 3 ($1 + 2 = 3$).

4. The reaction is zero order with respect to A and second order with respect to B. The overall order of reaction is 2 ($0 + 2 = 2$).

Test your understanding of concentration–time graphs and rate–time graphs by dragging the correct labels onto the graphs in **Interactive 1**. Start by labelling the axes.

Concentration

Time

Rate

Concentration

First order

Zero order

Second order

✓ Check

Interactive 1. What order?

More information for interactive 1

An interactive feature illustrates two graphs to help users understand reaction order. The first graph (on the left of the interface) plots concentration on the y-axis and time on the x-axis. The second graph (on the right of the interface) plots rate on the y-axis and concentration on the x-axis.

The first graph exhibits two concave upward decreasing curves, one decreasing more steeply than the other, and a straight line with a negative gradient.

The second graph features a straight line passing through the origin with a positive gradient, a concave upward increasing curve, and a horizontal, stable line in the center.

Users are required to determine the reaction order: zero, first, or second. They must then drag and drop the appropriate labels onto the corresponding curves in both graphs.

This interactivity helps users to distinguish between zero-, first-, and second-order reactions using concentration versus time and rate versus concentration graphs.

Read below for the solution:

First graph (on the left of the interface):

More steeply decreasing concave curve - second order

Less steeply decreasing concave curve - first order

A straight line with a negative gradient - zero order

Second graph (on the right of the interface):

A straight line passing through the origin with a positive gradient - first order

Concave upward increasing curve - second order

Horizontal, stable line - zero order

This activity helps users learn how to recognize different reaction orders by analyzing the shapes of concentration-time and rate-concentration graphs. It provides an interactive way to understand the relationship between reaction rate and concentration.

Orders of reaction and reaction mechanisms

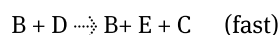
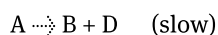
In [section R2.2.9 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-expression-and-its-experimental-id-45511/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-expression-and-its-experimental-id-45511/) you learned that the rate expression can only be determined experimentally. Although we know the molecularity of the overall reaction and can predict how many particles are colliding, why can we not predict their effect on the rate from the stoichiometry of the reaction?

It so happens that the rate expression cannot be deduced from the stoichiometry of the reaction. This only applies to overall reactions, and not elementary steps. Let's see how the order of reactions is related to the elementary steps in a reaction mechanism.

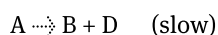
Cases where the slow step is the first step in the reaction mechanism

Overall reaction $2A \rightarrow 4B + C$

The order of this reaction was experimentally determined to be first order. The overall order will depend on the rate law from the rate determining step in the reaction mechanism. Since the order of the elementary steps can be determined from the reaction coefficients, we need to examine the proposed mechanism. The proposed mechanism for this reaction is:



Because the rate determining step is an elementary step, its rate law comes from its molecularity. Collision theory states that the concentration of each reactant in the rate determining step must appear in the rate law for that step, raised to the power of its stoichiometric coefficient in the equation. The slow step is:



The molecularity of the rate-determining step is one. That is why the rate law for this step is:

$$\text{rate} = k[A]$$

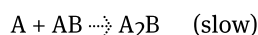
Table 1. The relationship between the molecularity of elementary steps and their rate law

Elementary step	Molecularity	Rate law from the elementary step
$A \rightarrow \text{products}$	unimolecular	$\text{rate} = k[A]$
$2A \rightarrow \text{products}$	bimolecular	$\text{rate} = k[A]^2$
$A + B \rightarrow \text{products}$	bimolecular	$\text{rate} = k[A][B]$

The rate law for the rate determining step leads us to the rate expression for the overall reaction. If the rate determining step is the first step in the reaction mechanism or the only step in a single step mechanism, then its rate expression is the same as the rate expression for the overall reaction.

Cases where the slow step is not the first step in the reaction mechanism

If the rate determining step is not the first step in the reaction mechanism, then deducing the reaction mechanism becomes a little more complicated. When the rate determining step comes after initial elementary steps the concentration of the other species in the previous steps need to be considered, as they also influence the rate. Consider a reaction between A and B:



The slow step is the second step. The rate of reaction will be governed by the slow step, so the rate expression will look like this:

$$\text{rate} = k[A][AB]$$

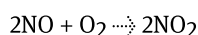
Except of course AB is not one of the reactants we started with. AB here is an intermediate. The rate expression is everything up to and including the slow step, except for any reaction intermediates. The concentration of AB must be expressed in terms of [A] and [B]. We can see that from the first equation that AB is formed from A + B. So substituting this into the equation instead of [AB] we get:

$$\begin{aligned}\text{rate} &= k[A][A][B] \\ &= [A]^2[B]\end{aligned}$$

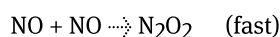
So the reaction is second order with respect to A and first order with respect to B.

Worked example 2

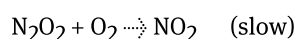
For the reaction:



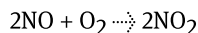
Step 1:



Step 2:



Step 2 is the rate determining step.

Overall:

Write the rate expression for the overall equation.

What is the order of the overall reaction?

The rate depends on step 2 for which the rate expression is:

$$\text{rate} = k [\text{N}_2\text{O}_2][\text{O}_2]$$

But N_2O_2 is not a reactant in the overall equation. It is a product of step 1 that is formed by $\text{NO} + \text{NO}$, that is, $[\text{NO}]^2$.

Substituting this into the rate equation gives:

$$\text{rate} = k [\text{NO}]^2[\text{O}_2]$$

Hence this is a third order reaction ($2 + 1 = 3$).

Aspect:

- Evidence
- Falsification

More than one reaction mechanism can be proposed for a chemical reaction. We cannot prove which mechanism is definitely correct for that reaction unless we can support it with evidence. On the other hand we can eliminate a mechanism if it is proven wrong or falsified. In the past where advancements in techniques like spectroscopy have led to the identification and quantification of a new intermediate, certain proposed mechanisms have been falsified.

Units of k vary depending on the order of reaction

Table 2 shows how the units of k can be calculated from the rate expression when the concentration of the reactants and the corresponding rate is known. It can be seen that the units of k depend on the reaction order.

Calculating units of k for a first order reaction we have:

$$\text{rate} = k[\text{A}]$$

$$k = \frac{\text{rate}}{\text{A}}$$

Evaluating the units gives:

$$\begin{aligned} k &= \frac{\text{mol dm}^{-3} \text{s}^{-1}}{\text{mol dm}^{-3}} \\ &= \text{s}^{-1} \end{aligned}$$

The units of k for zero order, second order and third order reaction can be calculated similarly as shown in **Table 2**.

Table 2. Calculating the units of k .

Zero order	First order	Second order
rate = k	rate = $k[A]$	rate = $k[A]^2$
$k = \text{rate}$	$k = \frac{\text{rate}}{[A]}$	$k = \frac{\text{rate}}{[A]^2}$
Substituting for units	Substituting for units	Substituting for u
$k = \text{mol dm}^{-3} \text{ s}^{-1}$	$k = \frac{\text{mol dm}^{-3} \text{ s}^{-1}}{\text{mol dm}^{-3}}$	$k = \frac{\text{mol dm}^{-3} \text{ s}}{(\text{mol dm}^{-3})^2}$
	$k = \text{s}^{-1}$	$k = \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$

Worked example 3

A reaction involving A and B is found to be first order with respect to A and zero order with respect to B.

The rate of reaction is $2.50 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$, when the initial concentrations of [A] and [B] are both $0.800 \times 10^{-3} \text{ mol dm}^{-3}$.

Calculate the value of the rate constant including its units.

The rate expression for this reaction is rate = $k[A]$.

Substituting the values into the rate expression gives:

$$\begin{aligned}
 2.50 \times 10^{-3} &= k \times 0.800 \times 10^{-3} \\
 k &= \frac{2.50 \times 10^{-3}}{0.800 \times 10^{-3}} \\
 &= 3.125 \\
 &= 3.13 \text{ (3 s.f.)}
 \end{aligned}$$

Substituting the units into the rate expression gives:

$$\begin{aligned}
 \text{mol dm}^{-3} \text{ s}^{-1} &= k \times \text{mol dm}^{-3} \\
 k &= \frac{\text{mol dm}^{-3} \text{ s}^{-1}}{\text{mol dm}^{-3}} \\
 &= \text{s}^{-1}
 \end{aligned}$$

Hence the rate constant is 3.13 s^{-1} (3 s.f.).

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer

- Thinkers
- Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
 - Social skills — Working collaboratively to achieve a common goal.
- **Time required to complete activity:** 40 minutes
- **Activity type:** Pair and individual activity

Activity 1

In this activity you will run a simulation in pairs. This is a half-life simulation.

Half-life is the time taken for the original concentration of the reactant to fall to half its original value. Radioisotopes follow this natural decomposition process where the time taken for the concentration of the radioisotope to fall to half is constant each time it decomposes.

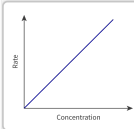
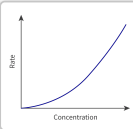
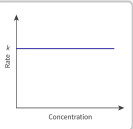
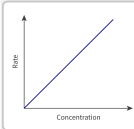
Open the simulation <https://academo.org/demos/radioactive-decay-simulator/>  (<https://academo.org/demos/radioactive-decay-simulator/>) and work through the following steps:

1. Start with 1600 atoms. Press the reset button. Enter a half-life of 20 in the space above. Enter 250 steps to be shown on the graph. Start the simulation by pressing the 'advance one step' button. Press the button 20 times to denote 20 steps.
 2. Record the number of atoms remaining after 20 steps. Round off to the nearest hundred.
 3. Next record the number of atoms remaining after another 20 steps. How are the number of atoms changing after each half-life of 20 steps?
 4. Continue pressing the button until the number of atoms remaining becomes zero. Describe the shape of the curve obtained. Which order of kinetics does this represent?
 5. Reset the simulation, this time enter a half-life of 40 steps. Tick the 'automatically advance steps' option and observe the graph formed. Describe the shape of this graph.
-
2. The number of atoms remaining after 20 steps is approximately 800.
 3. After a further 20 steps the total number of atoms remaining becomes approximately 400 hundred. After each half -life of 20 steps, the original number of atoms is halved.
 4. The graph is a curve with negative slope where the number of undecayed atoms halves after a fixed interval (half-life) of 20 steps, throughout the 250 steps. This represents first order kinetics.

5. The graph is similar in shape to D except it extends further as half-life is now increased.

Activity 2

Work individually to drag the cards under the right heading in **Interactive 2**.

Zero order	First order	Second order
Concentration–time graph		
Rate–concentration graph		
Rate = k		
Rate = $k[A]$	Rate = $k[A]^2$	Rate = k
		
		
		

 Check

Interactive 2. Features of orders of reaction.

 More information for interactive 2

An interactive table illustrates features of orders of reaction with missing fields. Users need to drag and drop the correct concentration-time graph, rate-concentration graph, and rate law expression ($\text{rate} = k$) for zero-order, first-order, and second-order reactions.

Drag and drop labels and graphs provided below the interface are;

$$\text{Rate} = k[A]$$

$$\text{Rate} = k[A]^2$$

$$\text{Rate} = k$$

A rate-concentration graph displays a straight line with a negative gradient.

A concentration-time graph displays a more steeply concave upward decreasing curve.

A rate-concentration graph displays a concave upward increasing curve.

A concentration-time graph displays a straight line passing through the origin with a positive gradient.

A concentration-time graph displays a less steeply concave upward decreasing curve.

A rate k-concentration graph displays a horizontal, stable line.

After completing the table, users can click the "Check" button to confirm whether their answers are correct and receive immediate feedback. This interactive table helps users to recognize how reaction order influences the shape of concentration-time and rate-concentration curves and apply rate laws.

Read below for the solution:

Zero order:

Concentration-time graph: Linearly decreasing curve

Rate-concentration graph: Horizontal, stable line

Rate law: $\text{Rate} = k$

First order:

Concentration-time graph: Less steeply concave upward decreasing curve

Rate-concentration graph: Straight line passing through the origin with a positive gradient.

Rate law: $\text{Rate} = k[A]$

Second order:

Concentration-time graph: More steeply concave with upward decreasing curve

Rate-concentration graph: Concave upward increasing curve

Rate law: $\text{Rate} = k[A]^2$